When Execution Composition Changes Trace Similarity

A Matched Evaluation of Behavioral Fingerprints

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Abstract

We study how AI-Agent participation changes errors in linking traces from the same benchmark task and whether behavioral fingerprints improve these decisions. In a five-share experiment with 17 CTH task groups, action-bigram cosine loses 10.92 percentage points in balanced accuracy (BA) from 0% to 100% Agent roles as missed links increase and false merges decrease. The five-share mean BA changes from adding local behavior and sequence features (B/S) are uncertain for all three classifiers. In a frozen external evaluation of 160 NYU D-CIPHER Agent-only challenge groups, B/S raises BA by 5.71 points for XGBoost (95% paired interval [3.63, 7.73]) and 3.64 points for the decision tree ([1.97, 5.37]). Both models miss fewer same-challenge links and falsely merge more different-challenge pairs. For XGBoost, these error rates change by −26.10 and +14.67 points, respectively. SVM loses 8.23 points of BA on the same pairs. On 109 AIT troubleshooting runs, the association gain from observed timing (T) remains uncertain.

1 Introduction

Security investigations must decide which retained traces deserve joint inspection. Event-correlation systems use temporal, causal, provenance, or technique-level evidence to connect records into incident narratives [10, 12, 9, 11]. Many available traces, however, expose only commands, coarse actions, and sometimes timestamps. A pairwise method that uses those fields may be sensitive to how a task was executed: Human and Agent runs can follow different paths through the same task, while different tasks can share similar tool patterns.

This creates two distinct decision errors. A *false split* rejects two traces from the same benchmark task; a *false merge* accepts traces from different tasks. We first ask how these errors vary when the Agent-role share of a fixed candidate comparison rises from zero to one. We then test whether a log-derived B/S/T fingerprint reduces these errors when added to the same learned association models and candidate pairs. The relation label is benchmark task identity, used as a retrieval proxy. It is not verified attack-activity, campaign, or operator identity.

The exact-mixture experiment measures how errors from a fixed similarity threshold change across role allocations. The matched CTH experiment estimates the B/S increment on identical pairs for three learners, and AIT supplies observed launch times for the timing ablation. In the frozen NYU D-CIPHER evaluation, B/S raises BA for XGBoost and the decision tree by recovering more same-challenge links while accepting more different-challenge pairs. SVM loses BA on the same NYU pairs.

Our contributions are: (1) a coupled five-share mixture test that reports false merges and missed links separately; (2) a matched, three-classifier comparison and a single B/S/T ablation table across CTH and AIT; and (3) a one-shot, cross-source Agent-only confirmation with explicit error transitions. We state what each task-proxy label permits and retain both unfavorable and uncertain results.

2 Related work

Attack-correlation methods align event, provenance, and technique evidence to reconstruct or search for attack behavior [10, 12, 9, 11, 13]. Agent-trajectory work instead studies workflow representation or identification: Trace analyzes terminal behavior, LIDAR enrolls model–harness fingerprints, Known By Their Actions studies browser-agent traces, and ProcGrep treats coding trajectories as programs [1, 3, 4, 2]. These tasks motivate log-derived behavioral features, but model identity and same-task trace association are different prediction targets. Cross-Agent studies campaign linking with synthetic labels [6]; those labels also differ from the benchmark-task proxy used here. The experimental

tables in the appendix preserve the broader earlier data-coverage and source-specific comparisons without combining their protocols with the new matched study.

3 Representation and association model

Let x and y be two traces, and let $z = 1$ indicate that they share a benchmark task. The base evidence $E(x, y)$ contains command TF–IDF cosine and action-bigram TF–IDF cosine. Each trace also receives a windowed fingerprint: B summarizes local action organization and repetition; S summarizes novelty, revisits, and fragment reuse; T summarizes measured launch gaps or durations only when the source records the required time fields. Missing observations remain missing. No time value is inferred from event order or a workflow label.

For each available block, the pair representation appends the absolute difference between training-standardized trace summaries. In the selected conditional fusion, these differences are multiplied by command similarity before concatenation with E. B/S/T are evidence features for the pair classifier; the task, execution role, source, and path are not model inputs. The classifiers are RBF-SVM, a decision tree, and XGBoost. Their scores are not calibrated probabilities. If either trace lacks a complete required block, the pair uses the same-fold E-only score and threshold; these fallback decisions remain in the evaluation.

4 Evaluation

Exact CTH role mixture. The frozen CTH core has 17 task groups. In 170 coupled allocations, 16 tasks contribute query and candidate roles, with exactly 0, 8, 16, 24, or 32 Agent roles among the 32 allocated roles. Each query has one same-task and three different-task candidates. Candidate tasks and draws are paired across shares. The share is therefore a property of the constructed query galleries, not a resampled proportion of all source records. We retain the earlier command- and action-bigram-cosine baselines to describe this protocol.

Matched classifier and ablation tests. The main CTH comparison uses the same 54,400 candidate-pair decisions for E and E+B/S across all five shares and all three learners. AIT contains 109 independent troubleshooting runs across 12 tasks; each configuration uses the same 43,600 pair decisions, with T computed from recorded launch timestamps. NYU contains 585 runs from four snapshots of one Agent framework, grouped into 160 challenge tasks not used in CTH development. Every NYU query is tested against one same-challenge and three different-challenge candidates. This is an AA-only cross-source test, and a challenge filename is the task-retrieval proxy.

Preprocessing, representation choice, classifier parameters, and thresholds are selected within task-disjoint training folds. Each evidence arm selects its threshold on inner out-of-fold predictions under the FMR-at-most-10% rule; the selected threshold is fixed for the outer evaluation. This constraint is not expected to transfer to the test tasks. CTH results average task outcomes and then the five shares equally; AIT and NYU average task outcomes. We use 10,000 paired task-cluster bootstrap resamples for intervals. Repeated gallery draws and pair decisions within a task are not independent observations, so intervals describe task-cluster uncertainty and remain approximate under shared candidates. No results are adjusted for the many development analyses.

We report balanced accuracy (BA), false-merge rate (FMR), and false-split rate (FSR), where $\mathrm{BA} = 1 - (\mathrm{FMR} + \mathrm{FSR})/2$. Increasing one error can therefore be hidden by a reduction in the other; all three measures are shown.

5 Results

5.1 The mixture baseline shifts toward missed links

Under the exact CTH role mixture, command-cosine BA changes from 53.59% at 0% Agent roles to 46.50% at 100% (endpoint difference -7.10 pp, 95% task interval $[-15.45, +0.06]$). Action-bigram BA changes from 57.40% to 46.48% (-10.92 pp, $[-20.61, -3.06]$). Across these endpoints, FSR rises and FMR falls. Thus the observed overall decline reflects a shift in the fixed operating point toward rejecting links, not an increase in both error types. This pattern belongs to this task-proxy allocation and threshold protocol.

5.2 Matched CTH learners do not show a common B/S gain

Figure 1 shows the same-pair E and E+B/S curves. Averaged equally over the five shares, the XGBoost B/S arm changes BA by −2.38 pp (95% paired interval [−7.13, +2.25]), while FMR rises 9.46 points and FSR falls 4.70 points. SVM changes BA by +1.38 pp [−3.83, +6.53]; its mean FMR also rises. The decision-tree B/S version rejects every candidate, giving 50% BA, zero FMR, and 100% FSR. None of the three learners satisfies the pre-specified requirement of a stable gain without an error trade-off. The small, five-share changes are made explicit in Figure 2; the intervals cross zero.

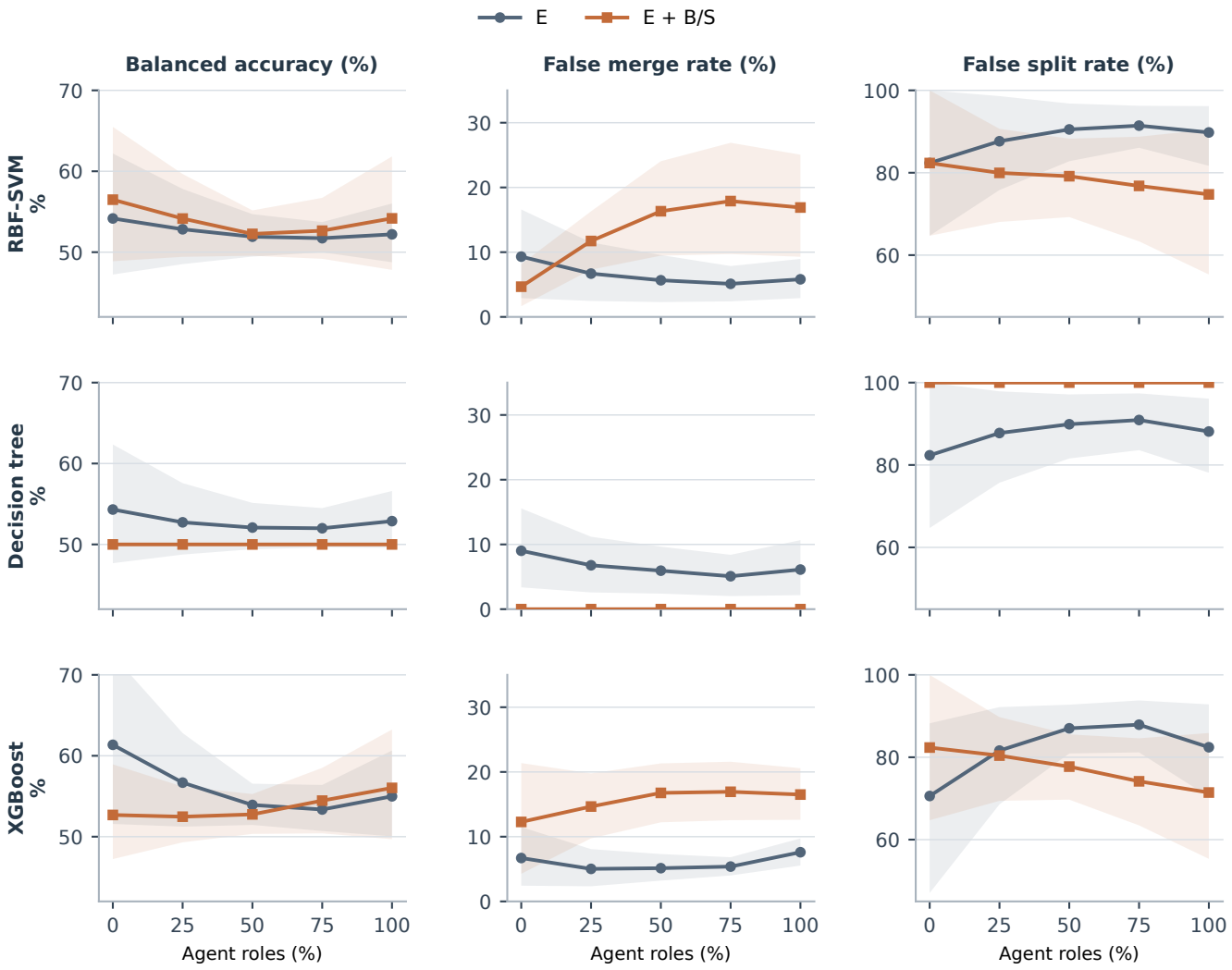


Figure 1: Matched CTH comparison over exact Agent-role shares. Rows are learners; columns are BA, FMR, and FSR. Lines show E and E+B/S, and bands show 95% task-cluster intervals. Y-ranges are shared within each metric column and stated by the axes. The role percentage is over query and candidate roles.

5.3 Pair type changes the error pattern, not the conclusion

We reuse the same XGBoost predictions and average only the shares where each candidate-pair type exists. For HH, the BA difference is −8.84 pp [−20.34, +1.67]; HA is +1.29 pp [−1.78, +4.71]; AA is +1.39 pp [−5.80, +8.03]. HA and AA point gains accompany higher FMR, while HH trends downward in BA. No stratum establishes a reliable gain. Pair types describe the modes assigned to the two traces and say nothing about whether their behaviors came from the same real attack or operator (Appendix C).

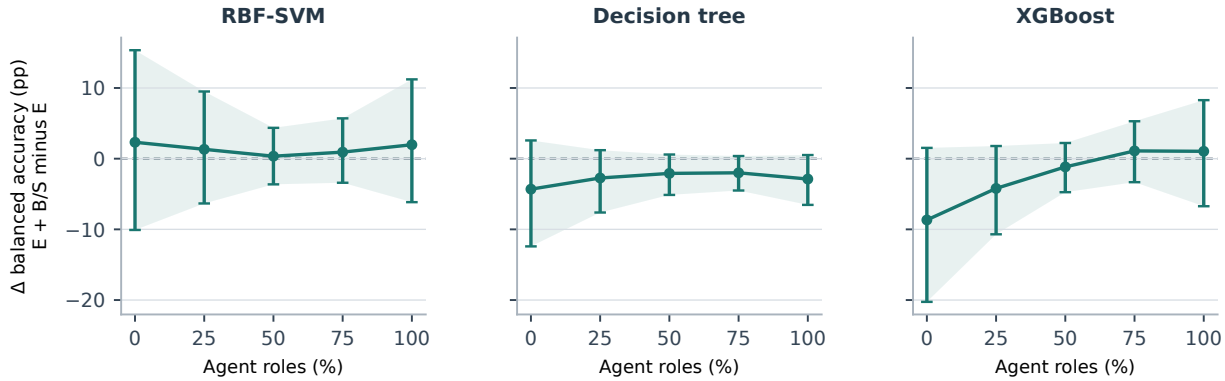


Figure 2: Paired BA change (E+B/S minus E) at each CTH share, in percentage points. The dashed line marks zero; intervals are task-cluster paired 95% intervals.

Table 1: Matched CTH classifier comparison averaged equally over five Agent-role shares. Every classifier uses the same paired candidate decisions. Changes are B/S minus evidence-only E in percentage points; 95% intervals are paired task-cluster bootstrap intervals.

Classifier	BA: E	BA: E+B/S	ΔBA [95% CI]	ΔFMR	ΔFSR
RBF-SVM	52.57	53.95	+1.38 [−3.83, +6.53]	+6.99	-9.74
Decision tree	52.80	50.00	−2.80 [−6.61, +0.57]	-6.59	+12.19
XGBoost	56.06	53.68	−2.38 [−7.13, +2.25]	+9.46	-4.70

5.4 AIT timing and external Agent-only transfer

On the AIT association task proxy, adding measured T to E+B/S changes BA by +0.98 pp [−0.21, +2.10]; the interval spans zero. The unified ablation in Table 2 shows that the point estimate is small and that removing B, S, or T does not identify a stable independent benefit across both datasets. AIT task and collection workflow co-vary with its execution labels, so the timing contrast is limited to this troubleshooting cohort.

On frozen NYU Agent-only challenge retrieval, adding B/S improves BA by +5.71 pp [+3.63, +7.73] for XGBoost and +3.64 pp [+1.97, +5.37] for the decision tree. With XGBoost, FSR falls from 86.96% to 60.86% and FMR rises from 10.72% to 25.39%. The tree shows the same error trade-off. SVM BA falls by 8.23 pp [−10.22, −6.28]. XGBoost also ranks the same-challenge candidate first more often, with a Hit@1 gain of +3.20 pp [+0.45, +5.90] (Appendix B).

6 Discussion

On NYU, the reduction in missed links outweighs the increase in false merges for XGBoost and the decision tree under BA's equal weighting of the two error rates. XGBoost's Hit@1 gain supports using its score to prioritize candidates for review on this benchmark. For automatic linking, the observed 25.39% FMR with B/S exceeds the 10% validation ceiling used to select the threshold on CTH. These two uses put different demands on the score and its decision threshold.

The HH/HA/AA breakdown shows why a single overall average is incomplete, but it is descriptive. Its share support and task candidates overlap, and there is no independent mixed-role confirmation dataset. The AIT timing block can be computed from real timestamps, yet its association increment is uncertain and its labels are entangled with task and collection workflow. These results do not identify what caused a particular pair to differ; similar Agent paths, changed execution paths, collection-interface differences, and threshold transfer remain plausible contributors.

All relations are benchmark-task proxies; independent attack-activity labels are absent. CTH terminal and Agent tool-call capture differ, and NYU's snapshots share one framework. Repeated draws do not add independent tasks. Because several alternatives were inspected during development, intervals are exploratory and unadjusted; operational linkage and attribution remain untested.

Table 2: Matched B/S/T ablation on the same candidate pairs. BA is task-macro balanced accuracy in percent; FMR and FSR are error rates in percent; ΔBA compares with the full fingerprint for that dataset. Brackets are 95% paired task-cluster bootstrap intervals (10,000 resamples). CTH averages the five role shares equally; AIT averages 12 tasks. Repeated pair decisions are not independent observations.

Configuration	B	S	T	BA (%) [95% CI]	FMR (%)	FSR (%)	ΔBA (pp) [95% CI]
CTH: 17 task groups; five exact role shares							
Full	✓	✓	–	53.68 [50.68, 56.96]	15.43	77.21	Reference
Remove B	–	✓	–	56.10 [51.99, 60.39]	14.47	73.34	+2.42 [−3.09, +8.05]
Remove S	✓	–	–	52.75 [49.67, 55.96]	13.13	81.37	−0.93 [−2.52, +0.59]
E only	–	–	–	56.06 [52.42, 59.91]	5.97	81.91	+2.38 [−2.28, +7.29]
AIT: 12 task groups; observed launch times							
Full	✓	✓	✓	51.92 [50.58, 53.18]	5.37	90.79	Reference
Remove B	–	✓	✓	52.01 [51.08, 53.03]	3.53	92.45	+0.09 [−1.05, +1.14]
Remove S	✓	–	✓	51.37 [50.15, 52.84]	5.24	92.02	−0.55 [−1.66, +0.51]
Remove T	✓	✓	–	50.94 [49.38, 52.58]	5.28	92.84	−0.98 [−2.10, +0.21]
E only	–	–	–	50.12 [50.00, 50.36]	0.00	99.76	−1.80 [−3.14, −0.37]

7 Conclusion

On external Agent-only challenge retrieval, B/S improves balanced accuracy for XGBoost and the decision tree, with fewer missed links and more false merges. XGBoost also improves the ranking of the correct candidate. SVM loses BA on the same pairs, and the five-share CTH experiment and AIT timing ablation leave their respective BA increments uncertain. For pair decisions, reporting both error rates alongside BA makes the cost of recovering more links visible.

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A Supplementary CTH and external results

Figure 3 displays paired changes by role type, while Table 5 gives exact XGBoost values by mixture share. The external table reports learner-specific absolute results and the paired error changes; its intervals cluster the 160 held-out challenge groups.

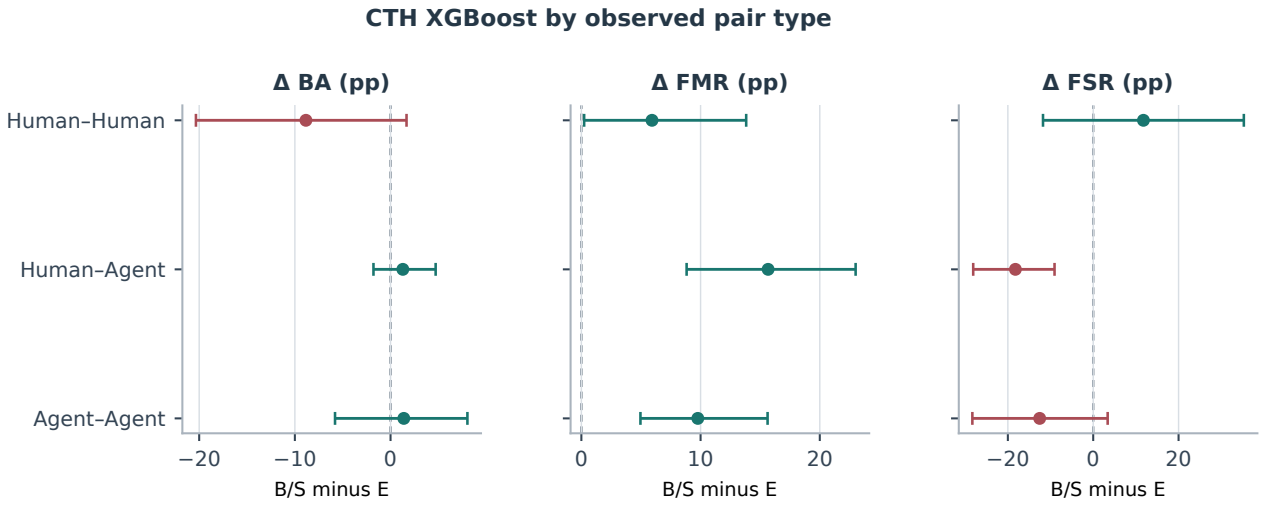


Figure 3: CTH XGBoost change (E+B/S minus E) by pair type and metric. Points are share-equal means over the eligible proportions; bars show paired task-cluster 95% intervals. HH, HA, and AA are evaluation roles, not attack identities.

Table 3: Frozen NYU D-CIPHER AA challenge-retrieval comparison. The challenge filename defines the same-task proxy. Changes are E+B/S minus E; intervals cluster by 160 task groups.

Classifier	BA: E	BA: E+B/S	ΔBA [95% CI]	FMR: E+B/S	ΔFMR	ΔFSR
RBF-SVM	58.16	49.93	−8.23 [−10.22, −6.28]	0.13	-26.45	+42.91
Decision tree	49.69	53.33	+3.64 [+1.97, +5.37]	8.54	+4.69	-11.98
XGBoost	51.16	56.87	+5.71 [+3.63, +7.73]	25.39	+14.67	-26.10

Table 4: CTH XGBoost error changes by pair type, averaged equally over the role shares where the type exists. HH, HA, and AA are trace-role pairs, not activity or actor labels.

Pair type	ΔBA [95% CI]	ΔFMR	ΔFSR
Human–Human	−8.84 [−20.34, +1.67]	+5.92	+11.76
Human–Agent	+1.29 [−1.78, +4.71]	+15.66	-18.23
Agent–Agent	+1.39 [−5.80, +8.03]	+9.77	-12.54

B Frozen external evaluation protocol

The NYU task manifest was frozen before fitting. Four public D-CIPHER snapshots from the official NYU-LLM-CTF leaderboard-submissions repository were included [15]. The exact source revision and hashes are in the frozen

Table 5: CTH XGBoost results by exact Agent-role share. Values are task-macro means over the same paired candidates; ΔBA intervals preserve task clustering.

Agent roles	BA: E	BA: E+B/S	ΔBA [95% CI]	FMR: E+B/S	FSR: E+B/S
0%	61.35	52.68	−8.67 [−20.26, +1.53]	12.28	82.35
25%	56.68	52.47	−4.21 [−10.70, +1.79]	14.66	80.40
50%	53.92	52.76	−1.16 [−4.74, +2.22]	16.76	77.72
75%	53.36	54.45	+1.10 [−3.33, +5.29]	16.94	74.15
100%	54.99	56.03	+1.04 [−6.73, +8.28]	16.51	71.43

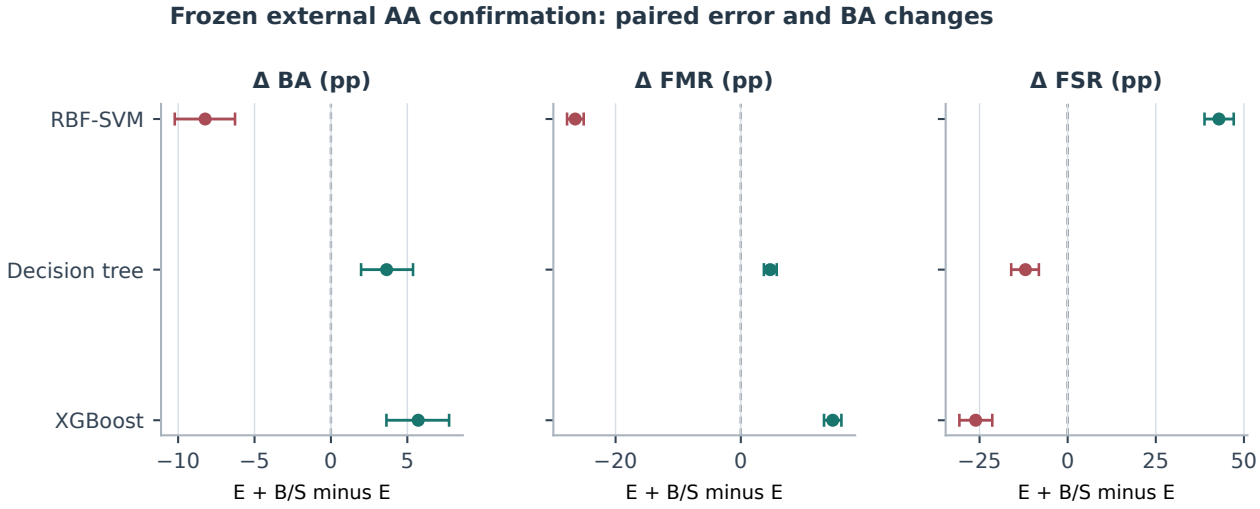


Figure 4: Frozen NYU AA evaluation: paired changes in BA, FMR, and FSR for E+B/S relative to E. The vertical dashed reference is zero; intervals cluster 160 challenge groups.

evaluation manifest. Challenge groups overlapping the CTH development pool and one duplicate command sequence were removed. The remaining 160 groups and 585 runs were partitioned by a fixed hash into three retrieval galleries. For each run query, 100 deterministic draws selected one other run from the same challenge and one run from each of three different challenges in the same partition. All three classifiers and both feature arms used the identical 234,000-pair list. Training used only the 17 CTH development task groups and all available training pairs; model parameters and thresholds were not selected from NYU outcomes. The CTH-derived threshold did not preserve its target FMR on the external source.

The official repository and snapshot metadata identify challenge filenames and run configuration. A shared challenge name is only the relation proxy. The four snapshots all belong to one Agent framework; this dataset cannot test Human–Agent associations or activity/actor attribution. Error transitions are counts of repeated pair decisions, not independent events: XGBoost corrected 11,241 earlier false merges and introduced 37,348 new false merges, while it corrected 19,708 earlier false splits and introduced 4,329 new false splits. These counts explain the rate changes but do not increase the independent sample size.

C Pair-type support and interpretation

The CTH type-specific estimates reuse the outer predictions. HH is supported at 0–75% Agent roles, HA at 25–75%, and AA at 25–100%. Each type is averaged equally across the shares where it exists and then across tasks. At 50%, the XGBoost gallery contributes 662 positive and 2,004 negative HH pair decisions, 1,396 positive and 4,161 negative HA decisions, and 662 positive and 1,995 negative AA decisions. Pair decisions share queries, candidate tasks, and task clusters; the counts are denominators, not independent sample sizes. The interval widths and overlapping task support preclude interpreting the three rows as independent replications.

D Historical tables and earlier protocols

The following tables are carried over verbatim from the prior manuscript and retain their original captions, labels, column definitions, typography, spacing, colors, and cell markers. Their protocols differ from the matched v2 evaluation; values are historical and are not pooled with the new results.

D.1 Earlier source and cohort tables

Table 6: Related work by research objective, evaluated scope, observed evidence, and intervention. Candidate sets and task domains differ; Cross-Agent uses synthetic campaign labels.

Work	Objective	Evaluated scope	Observation	Intervention
System identity and workflow				
Trace [1]	Model family	7 families; 3 scaffolds	Terminal commands	Active follow-on
ProcGrep [2]	Workflow; search	10 coding Agents	Coding actions	Passive
LIDAR [3]	Model identity	36 models; 2 harnesses	Coding traces	Active probes
Known By Their Actions [4]	Browser model	14 models; 4 sites	UI events; time	Passive
Attack behavior and relation evidence				
CyBiasBench [5]	Attack preference	5 Agents; 3 targets	HTTP requests	Task conditions
Cross-Agent [6]	Campaign links	Synthetic campaigns	Events; text; time	Logs and probes
This study	Trace association	3 sources; 2 protocols	Command sequences	Passive

Table 7: Cohort coverage. Green robot and blue person icons mark recorded Agent and Human counts; absent classes are omitted. Counts refer to records or sessions; external-source analyses are separate from the downloadable registry.

Cohort	Records	Mode counts	Observable evidence	Task or evaluation
Packaged registry	1,407	1,063 + 344	Command view	Catalog; not pooled
Original CTF	97	40 + 57	Commands; artifacts	40 tasks; fixed pair panel
CTH mixed-mode	175	118 + 57	Command order	40 tasks; cross-mode retrieval
CTH mixture core	85 raw	51 + 34	Command order	17 tasks; role mixtures
Antieval	163	163	Command order	32 tasks; Agent retrieval
KYPO	197	197	Command order	97 people; task-level pairs
AIT audit logs	109	48 + 61	Launches; timing	12 tasks; mode recognition
Honey [7]	13,080	13,080	Commands; duration	Same prompt/environment condition
CyBiasBench [5]	450	450	HTTP method; route	Target and requested family

Table 8: Observed fields and evaluation role by source. A check marks an observed field; a dash marks an unavailable field. Honey execution duration is not treated as an event timestamp.

Source	Unit	Order	Start time	Duration	Modes	Evaluation role
CTH mixed tasks	175 traces	✓	–	–	A + H	Same-task retrieval; exact role mixtures
AIT audit logs	109 runs	✓	✓	–	A + H	Mode recognition; observed launch rhythm
Honey for the Agent	13,080 sessions	✓	–	✓	A	External-source command and duration analysis
CyBiasBench	450 sessions	✓	–	–	A	Target-held-out request retrieval

D.2 Earlier association and ablation tables

Table 9: Complete source-held-out association comparison. Training uses the other sources; B/S windows contain eight events. BA is in percent; differences and unadjusted 95% cluster intervals are in percentage points. Green/red encode improvement/decline in the point estimate; interval bounds retain the uncertainty.

Test source	Evidence	+ B/S	ΔBA [95% CI]
CTH	51.98	56.69	+4.71 [+0.69, +8.83]
KYPO	66.40	66.61	+0.21 [−1.06, +1.36]
Antieval	71.90	68.63	−3.27 [−6.28, −0.22]

Table 10: CTH fingerprint ablation. BA is task-macro balanced accuracy (%); Δ compares each row with the full B/S model in percentage points. Intervals resample the 40 paired task-level outcomes 10,000 times; they are exploratory and unadjusted for multiple comparisons.

Configuration	B	S	BA (%)	Δ vs. full [95% CI]
Evidence only	–	–	65.07	−2.03 [−3.98, −0.08]
Evidence + length	–	–	65.78	−1.32 [−4.12, +1.49]
Remove B	–	✓	63.96	−3.14 [−6.21, −0.22]
Remove S	✓	–	65.63	−1.46 [−4.21, +1.07]
Full B/S	✓	✓	**67.10**	Reference

Table 11: AIT feature controls, 109 runs. BA (%) and 95% task-cluster intervals are shown for both holdouts. Shuffled controls average 20 perturbations per trace.

Configuration	Task-group BA [95% CI]	Source-group BA [95% CI]
B+S, full trace	53.0 [41.0, 65.5]	35.8 [29.9, 42.4]
B+S, first eight events	72.3 [64.9, 79.0]	41.4 [33.7, 48.4]
B+S, window order shuffled	55.8 [49.7, 61.7]	40.3 [31.9, 48.6]
B+S+T, full trace	68.8 [61.8, 75.8]	52.6 [43.4, 62.7]
B+S+T, gap order shuffled	70.6 [61.9, 79.5]	49.9 [44.1, 55.9]

D.3 Earlier feature, transfer, and recognition tables

Table 12: Window-level feature definitions. A value is missing when its required observations are absent or insufficient.

Block	ID	Measurement
B	B1–B3	Distinct command share; Shannon entropy of command names; share of the dominant command.
B	B4–B6	Fraction of known adjacent action pairs that switch group; conditional entropy of the next action; longest single-action run divided by window length.
B	B7–B8	One minus the unique mapped adjacent-action-pair share; fraction of adjacent command names that are identical.
S	S1–S3	First-seen command share; fraction of repeated commands whose previous occurrence is at least two positions back; mean normalized log revisit span.
S	S4–S5	A-B-A action return share among valid triplets; share of valid four-action segments whose first and fourth action match.
S	S6–S8	Reused adjacent command-bigram share; reused command-trigram share; Jensen–Shannon divergence between current and preceding window action distributions.
T	T1–T3	Median log(1+gap); interquartile range of log gaps; burstiness $(\sigma - \mu)/(\sigma + \mu)$ over nonnegative raw gaps.
T	T4–T6	Fraction of gaps at most one second; fraction at least ten seconds; lag-one correlation of log gaps when at least three valid gaps exist.
T	T7–T8	Median log(1+observed event duration); median log(1+idle interval) after subtracting observed duration from the next start time.

Table 13: Additional transfer comparisons. Each row uses its stated outcome and held-out unit; changes are percentage points. The CyBiasBench interval is the range across three targets, not a confidence interval. Legacy denotes the earlier 24-value profile.

Source / n	Holdout	Comparison; outcome	Δ [95% CI]
CTH / 175	Task; opposite mode	Command → +B/S; Hit@1	-0.08 $[-5.71, +5.58]$
CTH / 219	Task; Agent-only	Command → legacy; Hit@1	-3.57 $[-6.82, -0.43]$
AIT / 109	Task	B/S → B/S/T; BA	$+15.83$ $[+6.11, +25.00]$
AIT / 109	Source group	B/S → B/S/T; BA	$+16.88$ $[+8.19, +25.77]$
Honey / 13,080	Model snapshot	Command → B/S; Hit@1	-1.30 $[-2.21, -0.35]$
CyBias / 450	Target	Route → +B/S; Hit@1	$+0.92$; $[-0.17, +1.58]^{\dagger}$
KYPO / 197	Participant	Command → +B/S; Hit@1	-1.42 $[-2.59, -0.23]$
Antieval / 120	Task	Command → legacy; Hit@1	-2.89 $[-7.50, +1.79]$

Table 14: Frozen 89-run AIT readouts: task-macro BA (%) and 95% task-cluster intervals.

Readout	Task-group holdout	Source-group holdout
Length	46.6 [32.4, 60.1]	39.5 [22.7, 57.2]
Legacy K1–K8	55.2 [45.6, 66.2]	44.9 [33.4, 56.1]
B	56.3 [40.9, 71.6]	39.9 [25.6, 54.2]
B+S	55.5 [45.6, 65.6]	39.6 [28.5, 50.8]
B+S+T (linear)	70.6 [63.6, 78.5]	44.4 [31.7, 54.7]
B+S+T (trees)	73.0 [63.1, 82.4]	44.9 [32.8, 56.7]
Command TF–IDF	96.8 [94.3, 99.2]	82.6 [69.4, 92.8]

E Reproduction record

The research package retains the v1/v2 evaluation scripts, frozen manifests, predictions, verification logs, and bounded development comparisons. The raw AIT archive and original run commands are not redistributed. The source bundle contains the permitted analysis materials and every figure referenced here. Source-specific upstream terms apply.

Table 15: Post-observation refinement on the identical 40-task, 175-query CTH galleries. Model family and regularization selection use inner tasks. All point estimates and paired task-cluster intervals are retained.

Method	Hit@1 (%)	MRR	ΔHit@1 [95% interval], pp
Command cosine	33.65	.539	Reference
Scalar fusion, inner-selected C	34.19	.540	+0.54 [−4.67, +5.88]
Conditional B/S fusion	30.02	.518	−3.63 [−9.13, +1.29]
48 component differences	23.92	.453	−9.73 [−19.00, −0.87]
Local masked DTW fusion	32.44	.530	−1.21 [−7.54, +5.08]
Nested family selector	33.35	.537	−0.29 [−5.67, +5.21]
Command + cross-fitted mode	30.52	.518	−3.13 [−9.17, +2.21]
Command+B/S + cross-fitted mode	31.15	.522	−2.50 [−9.13, +3.67]